HealthConnect Final Model & Data Science Report

Week 8 - Final Integration, Documentation & Presentation

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1. Executive summary

Week 8 finalizes the HealthConnect Data Science contribution by selecting and documenting the strongest defensible admission-modelling configuration produced after Week 7 testing and cross-track validation. The final documented candidate is a leakage-safer Gradient Boosting pipeline evaluated on 5,000 integrated clinical appointments (586 admissions; 4,414 non-admissions). It achieved holdout ROC-AUC 0.5189 and 5-fold cross-validated ROC-AUC 0.5237 +/- 0.0246. These values remain close to chance. The final decision is therefore not to deploy the model.

2. Week 7 to Week 8 transition

Transition item	Week 8 response
Main Week 7 output	Model testing, refinement and cross-track feature validation package
Most important result	Model discrimination remained close to chance under holdout and cross-validation
Major issue	Weak predictive signal plus prediction-time leakage risk for Diagnosis/Severity
Refinement	Leakage-safer pre-assessment feature set informed by Data Analytics
Ready component	Final documented model pipeline, evaluation, feature decisions and risk statement
Still needed for deployment	Stronger pre-admission predictors, fresh untouched test set and operational target definition
Final integration track	Data Analytics (Agu Precious); ML Engineering handoff requirements documented
Presentation focus	Why responsible validation led to a do-not-deploy decision

3. Final component and intended purpose

The final Data Science component is a reproducible binary classification pipeline for hospital admission status. The intended prediction point is patient arrival, before initial clinical assessment. This timing constraint is important because a valid operational model cannot depend on information recorded after the decision point.

4. Data and target

- Integrated sample: 5,000 appointment-level records.
- Target: actual hospital admission status (`Admitted`) derived from the Admissions sheet linked by AppointmentID.
- Class distribution: 586 admitted (11.72%) and 4,414 not admitted (88.28%).
- No attendance/no-show field exists in the supplied Rwanda appointment data; attendance prediction is therefore not claimed.

5. Final feature and modelling decisions

Feature	Decision	Reason
AgeAtAppointment	Retain	Calculated as of appointment date
Gender	Retain	Supported by Data Analytics feature review
HasChronicCondition	Retain	Supported by Data Analytics feature review

VisitType	Retain with caveat	Supported with caveat
FacilityID	Retain with caveat	Rating differences were not significant, but ID predictive value is a separate question
DepartmentID	Retain with caveat	Department rating differences not significant; ID may still provide context
PriorVisits	Add	Suggested by Data Analytics; calculated strictly from earlier appointments
AppointmentMonth	Add	Suggested time/context feature
AppointmentWeekday	Add	Suggested time/context feature
Diagnosis	Exclude from pre-assessment model	Potential prediction-time leakage
Severity	Exclude from pre-assessment model	Potential prediction-time leakage
MaritalStatus	Exclude from final cross-track model	Not validated in analytics review
Education	Exclude from final cross-track model	Not validated in analytics review
BloodGroup	Exclude from final cross-track model	Not validated in analytics review
BMI	Exclude from final cross-track model	Not validated in analytics review

6. Final evaluation protocol

The final cross-track configuration uses a stratified 80/20 split with random seed 42 for comparability with Week 7 and 5-fold stratified cross-validation as the stronger generalisation evidence. The 20% holdout contains 117 positive and 883 negative cases. Because this partition had already been inspected during Week 7, it is not described as a pristine one-time test set; this limitation is explicitly documented.

7. Model evaluation

Model	Holdout ROC-AUC	PR-AUC	Accuracy	Precision	Recall	F1	CV ROC-AUC
Final documented candidate - Gradient Boosting	0.5189	0.1199	0.882	0.000	0.000	0.000	0.5237 +/- 0.0246
Balanced Logistic Regression comparator	0.5024	0.1154	0.538	0.116	0.444	0.184	0.4986 +/- 0.0322

Gradient Boosting is retained as the final documented candidate because it has the best verified discrimination in the leakage-safer configuration. This is a documentation/model-selection decision, not a deployment approval.

8. Baseline vs final model

Stage	Model	Holdout ROC-AUC	CV ROC-AUC	Decision
Week 6 candidate	Gradient Boosting	0.4967	N/A	Historical candidate; near chance
Week 7 tested candidate	Gradient Boosting	0.4917	0.4811	Rejected for deployment
Week 7 threshold/refinement comparator	Refined comparator	0.5420	0.4875	Rejected as unstable
Week 8 final documented candidate	Gradient Boosting	0.5189	0.5237	Final documented candidate; NOT deployment-ready

9. Error analysis summary

At the default 0.50 threshold, the final Gradient Boosting candidate produced TN=882, FP=1, FN=117 and TP=0 on the 1,000-record holdout set. The model therefore missed every admitted case at this threshold. Accuracy (0.882) is misleading because the negative class dominates. ROC-AUC, PR-AUC, balanced accuracy and recall are more informative for this task.

10. Cross-track contribution

Agu Precious (Data Analytics) supplied validated feature and data-quality findings. These findings directly changed the Data Science feature set: Diagnosis and Severity were held out due to timing/leakage concerns; Age

was aligned to appointment date; time-safe PriorVisits, month and weekday were added; FacilityID and DepartmentID were retained with caveats. Re-testing improved methodological validity but did not produce a reliable model. This is a meaningful integration outcome because the collaborator output changed the model design and interpretation.

11. Business interpretation and suitability

- The model cannot reliably discriminate which patients will be admitted.
- It must not be used for clinical triage, patient prioritization, staffing allocation, or automated care decisions.
- The current model may be retained only as a reproducible analytical/educational artifact.
- The do-not-deploy decision is itself the responsible business outcome of the validation process.

12. Limitations and risks

- Near-chance discrimination under holdout and cross-validation.
- Severe class imbalance: only 11.72% admitted.
- Holdout partition was inspected during Week 7 and is not a pristine final test set.
- No attendance/no-show target exists in the supplied appointment data.
- Some potentially useful predictors may not exist before the prediction point.
- Facility and department IDs should not be interpreted as quality rankings.
- The model artifact is not clinically validated or production-approved.

13. Final integration and ML Engineering handoff

A final technical handoff file is included for ML Engineering. It specifies the prediction point, input schema, preprocessing and output expectations, while explicitly stating the do-not-deploy constraint. If the HC-POD demonstrates pipeline compatibility, it must be labelled as a technical integration demonstration rather than operational deployment.

14. Final recommendations

- Collect a true attendance/no-show outcome if the project objective remains missed-appointment reduction.
- Obtain stronger pre-admission predictors that are genuinely available before clinical assessment.
- Freeze a fresh untouched test set before future model selection or threshold tuning.
- Define a business cost matrix for false negatives and false positives before selecting any operational threshold.
- Revisit deployment only after materially stronger and stable out-of-sample evidence.

15. Final conclusion

The Week 8 Data Science contribution is complete as a final documented modelling outcome. Cross-track validation improved the methodological safety of the feature set, but the available data did not support reliable admission prediction. The final candidate is therefore retained for reproducibility and presentation only, with a clear do-not-deploy recommendation. This conclusion is consistent with the project evidence and demonstrates evidence-based decision-making rather than forced model success.